Statistical Business Analysis Documentation

1. Project Overview

The Statistical Business Analysis project analyzes business sales data using statistical techniques. It evaluates sales trends, marketing effectiveness, hypothesis testing, confidence intervals, correlation analysis, and regression analysis to support data-driven business decisions.

2. Project Objectives

• Calculate descriptive statistics.
• Analyze data distribution.
• Perform hypothesis testing.
• Calculate confidence intervals.
• Analyze correlation between Marketing Spend and Total Sales.
• Build a Linear Regression model.
• Generate business insights and recommendations.

3. Setup and Installation Instructions

1. Install Python 3.10 or later.
2. Install required packages:

pip install -r requirements.txt

or

pip install pandas numpy matplotlib seaborn scipy scikit-learn statsmodels openpyxl

3. Place all project files in one folder:
- business_data.csv
- statistical_analysis.ipynb
- requirements.txt
- hypothesis_tests_results.txt
- statistical_report.pdf
- README.md

4. Open statistical_analysis.ipynb in VS Code or Jupyter Notebook.
5. Run all notebook cells in sequence.

4. Code Structure Explanation

• Cell 1: Import libraries
• Cell 2: Load dataset
• Cell 3: Data preprocessing

• Cell 4: Descriptive statistics
• Cell 5: Histogram and Box Plot
• Cell 6: Normality test
• Cell 7: Correlation matrix and Pearson correlation
• Cell 8: Heatmap
• Cell 9: Three hypothesis tests
• Cell 10: Confidence interval
• Cell 11: Linear regression
• Cell 12: Business insights and conclusion

5. Screenshots of Working Application

Include screenshots of:
• Dataset Preview
• Descriptive Statistics Output
• Histogram
• Correlation Heatmap
• Hypothesis Test Results
• Regression Graph
• Final Statistical Analysis Report

6. Explanation of Technical Requirements

✓ Calculated descriptive statistics (Mean, Median, Mode, Variance, Standard Deviation)
✓ Created histogram and tested normal distribution using Shapiro-Wilk Test
✓ Performed Pearson correlation and generated a heatmap
✓ Performed three hypothesis tests:
- One-Sample T-Test
- Independent T-Test
- One-Way ANOVA
✓ Calculated 95% Confidence Interval and Margin of Error
✓ Performed Linear Regression and calculated R^2 Score
✓ Generated business insights and recommendations

Output:

```
         Date     Product  Quantity  Price Customer_ID Region  \
0  2026-01-01      Laptop         5  43602       C1001  South   
1  2026-01-02      Mobile         9  16919       C1002   West   
2  2026-01-03      Laptop         2  43622       C1003  South   
3  2026-01-04  Smartwatch         1   7465       C1004  South   
4  2026-01-05  Smartwatch         4   6718       C1005   East   

   Marketing_Spend  Total_Sales  
0             6012       218010  
1             3424       152271  
2             5582        87244  
3            11195         7465  
4             9359        26872  
<class 'pandas.DataFrame'>
RangeIndex: 120 entries, 0 to 119
Data columns (total 8 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   Date             120 non-null    str  
 1   Product          120 non-null    str  
 2   Quantity         120 non-null    int64
 3   Price            120 non-null    int64
 4   Customer_ID      120 non-null    str  
 5   Region           120 non-null    str  
 6   Marketing_Spend  120 non-null    int64
 7   Total_Sales      120 non-null    int64
dtypes: int64(4), str(4)
memory usage: 7.6 KB
None
```

```
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0  2026-01-01      Laptop         5  43602       C1001  South
1  2026-01-02      Mobile         9  16919       C1002   West
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 5   Region           120 non-null    str
 6   Marketing_Spend  120 non-null    int64
 7   Total_Sales      120 non-null    int64
dtypes: int64(4), str(4)
memory usage: 7.6 KB
None
```
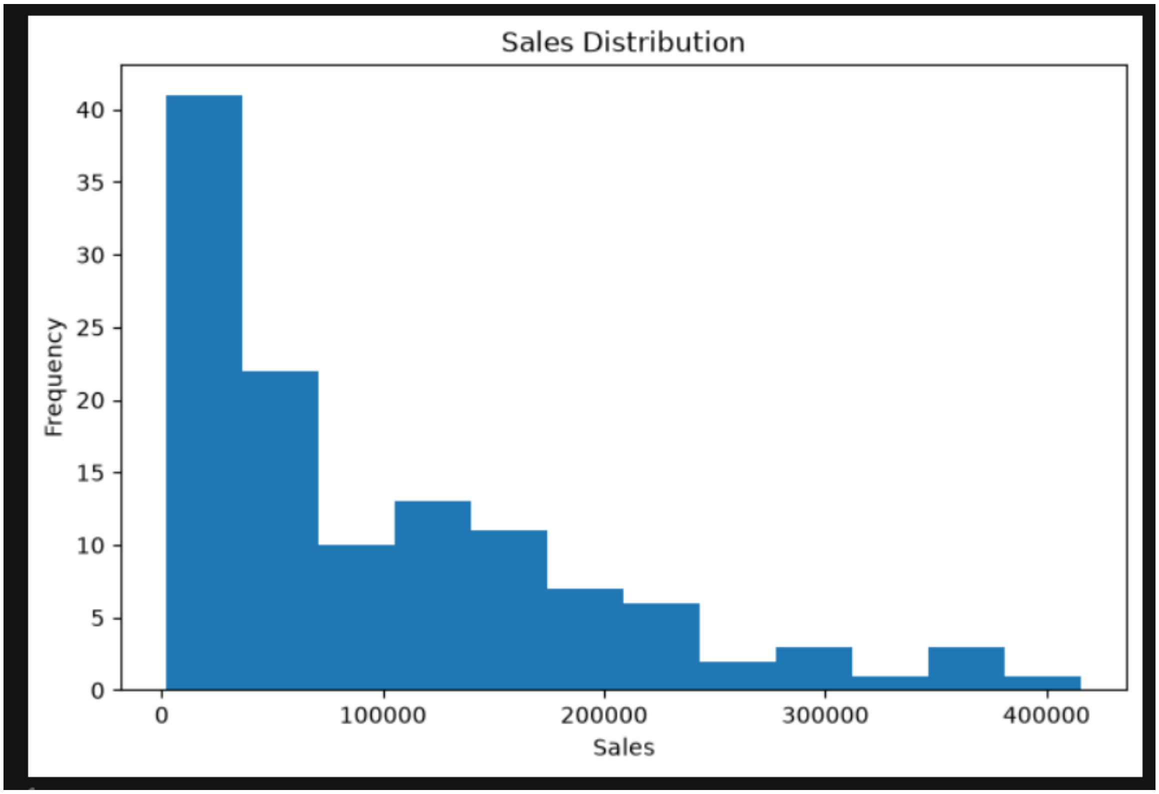
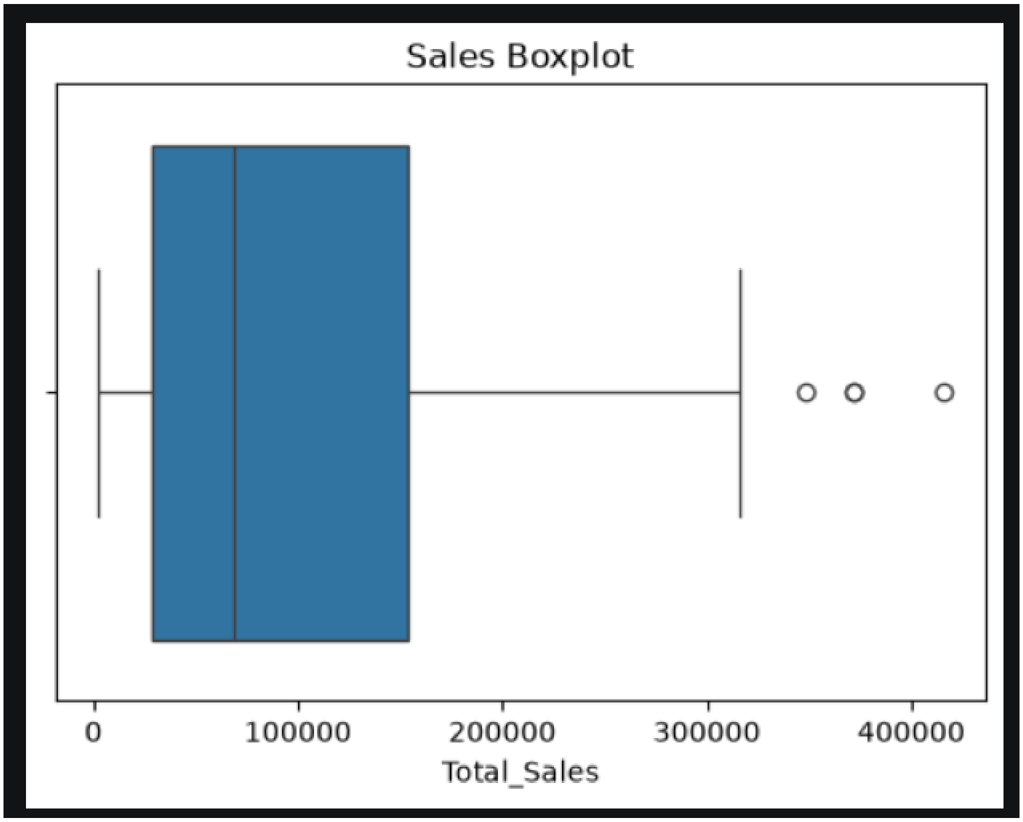

```
===== DESCRIPTIVE STATISTICS =====
         Quantity         Price  Marketing_Spend    Total_Sales
count  120.000000    120.000000       120.000000     120.000000
mean     5.691667  18747.741667      6961.358333  100933.416667
std      2.770783  14791.497693      2880.074937   94799.158480
min      1.000000   1024.000000      2009.000000    2158.000000
25%      4.000000   6290.000000      4598.750000   28672.000000
50%      6.000000  17515.500000      6868.000000   68321.000000
75%      8.000000  22273.000000      9411.250000  153502.500000
max     10.000000  46497.000000     11939.000000  415539.000000

Mean Sales = 100933.41666666667
Median Sales = 68321.0
Mode Sales =
0        2158
1        3244
2        3558
3        5020
4        6144
        ...
115    316393
116    348304
117    371904
118    371976
119    415539
Name: Total_Sales, Length: 120, dtype: int64
Variance = 8986880448.514006
Standard Deviation = 94799.15847998866
Minimum = 2158
Maximum = 415539
```


Sales Distribution
Frequency
Sales
0
5
10
15
20
25
30
35
40
0
100000
200000
300000
400000


Sales Boxplot
0
100000
200000
300000
400000
Total_Sales